

Mohamed El Baha

AI Research Engineer

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EXPERIENCE

Michelin | *Senior Data Scientist*

April 2022 - Present

- Develop deep learning and machine learning models for a wide array of use cases, including time series analysis and generative design.

- Analyze and preprocess various types of data to optimize model performance

Collaborate with cross-functional teams to integrate data-driven solutions into existing workflows

- Provide statistical training to Michelin employees to enhance their data analysis capabilities

Limsi | *Data Scientist*

May 2021 – Nov 2021

- Implemented Machine learning algorithms related to Eye Gaze movement and speech, to predict the score of English tests. Deployed these algorithms via a Python interface.

- Conducted speech analysis using Hidden Markov Models, utilizing forced alignment to accurately synchronize speech and transcription data, providing valuable insights into language patterns and usage.

Orano TN | *Engineering Intern*

Sep. 2020 – Dec 2020

- Directed a team of five engineering students in conducting a comprehensive benchmark study to evaluate and enhance the efficiency of engineering processes.

EDUCATION

M.Eng Mathematical and Computational Engineering

Top class 1%

Ecole des Mines de Nantes - Telecom Bretagne

Courses: Machine learning - Deep learning - Operational research - Optimization - Numerical Methods

Stochastic Dynamic Modeling - Data Science

B.Sc - Pure Mathematics

University of Western Brittany

Highly selective program to prepare entry Exams for engineering Schools

Courses: Calculus - Statistics - Complex Analysis - Algebra

SKILLS

Languages: English (*proficiency*), French (*Bilingual*), Arabic (*Native*)

Programming Languages: Python, Julia, R, Matlab

Tools & platforms: Git, GitLab, Kedro, Docker, Dataiku

Libraries & Frameworks Tensorflow, Keras, Pytorch, Pandas, HuggingFace, Numpy, Scikit, Seaborn

PUBLICATIONS

- *GeMOVA : Generative Models Visual Atlas*, **M.EL Baha**, F. Oubari

- *Multi-Component VAE with Gaussian Markov Random Field*, Fouad Oubari, Mohamed **El Baha**, AISTATS 2026

- *A System for Automatic Calculation of the Eye-Voice Span*, **M. El Baha**, O. Augereau, S. Kobylyanskaya, I. Vasilescu and L. Devillers, Document Analysis System 2021

- *Witt theory, a proof of witt equivalence theorem*, **M. El Baha**, T. Levasseur, LMBA 2019

SIDE ACTIVITIES

- Founder & President of MAIR : Moroccans in AI Research, the largest moroccan ressearcher in AI Community in Morocco.
- MoroccoAI Committee Executive: Non-profit Organization to Promote AI Growth in Morocco Across the Spectrum.
 - The Lead Organizer of InnovAI hackathon (+500 participants) December 2024.
 - The Lead Organizer of the GenAI hackathon (+150 participants) December 2023.
 - Mentor and organizer of AI Summer School (60+ participants) August 2021.
- Founder of SaveOurSeas

TEACHING

- Advanced Deep Learning - 5A : Polytech Clermont Ferrand - 2024-2025
- Statistics and data analysis for Performance Analysts : 2023-2024-2025

STUDENTS

- 2025 Shaimaa Abu Yousuf - Engineering Intern - Leukemia Detection Using Transfert Learning for Accurate Classification.
- 2024/2025 Rémy Vallot - PhD Candidate - ENS Paris Saclay - Solver Acceleration using ML methods
- 2023 Moad Yachouti - Engineering Intern - Université Paris Saclay - Explainable AI in Predictive ML models.

INTERESTS

AI Communities, Environment, Surf,